

CSCE 775: Deep Reinforcement Learning and Search

RL-Driven Parameter Optimization for Synthetic Marine Radar Augmentation via Echo Trails

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1 Introduction

Training robust object detectors for marine radar requires large, diverse labeled datasets, yet collecting and annotating real-world radar scans is expensive and constrained by geography, weather, and vessel availability. The EchoTrails framework addresses this by generating synthetic radar scenes with configurable targets, clutter, and a temporal augmentation called *echo trail rendering*, which composites historical frames with decay-weighted intensity to produce motion-encoded training images. A YOLOv8n detector trained on these augmented images learns to distinguish stationary from moving maritime targets based on their trail signatures.

However, the EchoTrails pipeline exposes a high-dimensional parameter space spanning trail rendering (trail length N , decay function, intensity encoding, color mapping), synthetic clutter generation (streak count, intensity range), and per-object-group configuration (speed, intensity profile, flicker behavior, placement). The existing ablation study [Cubuk et al., 2019] evaluates trail length in isolation while holding all other parameters fixed, leaving the joint optimization landscape unexplored. Manual grid search over this combinatorial space is prohibitively expensive, as each parameter configuration requires a full generation-training-evaluation cycle of approximately 2,500 frames and 50 YOLOv8n training epochs.

We propose a reinforcement learning framework that learns a policy to *jointly* optimize the EchoTrails generator parameters, maximizing downstream detection performance (mAP_{50-95}) on held-out validation data. By formulating parameter selection as a sequential decision-making problem and solving it with Soft Actor-Critic (SAC) [Haarnoja et al., 2018], our approach replaces exhaustive ablation with an adaptive, closed-loop search over the full parameter space. The agent discovers effective parameter configurations that no single-variable ablation would find, and can condition on scene characteristics such as object density and clutter presence to produce context-specific augmentation recipes.

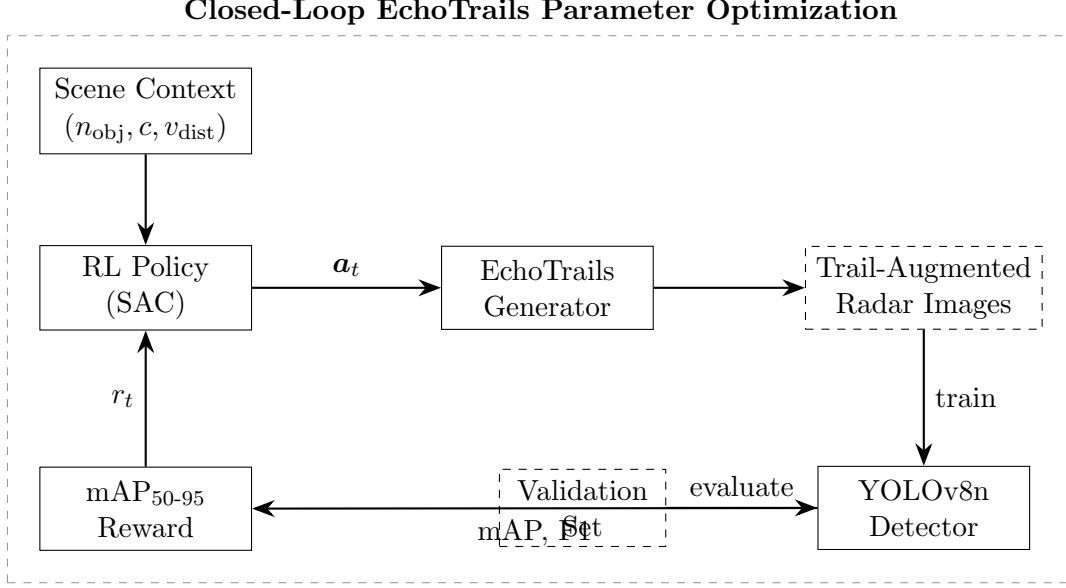


Figure 1: System architecture. The RL policy outputs EchoTrails generator parameters $\mathbf{a}_t$ conditioned on scene context (object count n_{obj} , clutter flag c , speed distribution v_{dist}). The generator produces trail-augmented radar images that train a YOLOv8n detector. Validation mAP₅₀₋₉₅ on a held-out set forms the reward signal.

2 Approach

We formulate EchoTrails parameter optimization as a Markov Decision Process (MDP). The state, action, and reward are defined over the actual generator parameter space exposed by the EchoTrails pipeline.

2.1 State Space

At each step t , the agent observes a state $\mathbf{s}_t = (\boldsymbol{\theta}_t, \mathbf{c}_t, \mathbf{m}_t)$ comprising three components. The current parameter vector $\boldsymbol{\theta}_t \in \mathbb{R}^k$ encodes the active generator settings. The scene context vector $\mathbf{c}_t$ encodes the composition of the current scenario, including total object count, the fraction of stationary versus moving targets, the speed tier distribution (slow at 2 px/frame, medium at 5 px/frame, fast at 12 px/frame), and the binary clutter flag. The metric history $\mathbf{m}_t$ contains recent validation scores (mAP₅₀₋₉₅, precision, recall) from the previous h evaluation rounds.

2.2 Action Space

The action $\mathbf{a}_t$ specifies a complete generator configuration drawn from the EchoTrails parameter space. The action vector comprises both continuous and categorical components.

The continuous parameters include trail length $N \in [0, 30]$ (number of history frames composited per output), clutter streak count $n_{\text{streaks}} \in [0, 150]$, clutter intensity bounds

$[I_{\min}, I_{\max}] \subset [0, 255]$, per-group base intensity $\beta \in [0.2, 2.0]$, intensity variability $\sigma \in [0.0, 0.5]$, and flicker rate $\rho \in [0.0, 1.0]$ with associated intensity drop $\delta \in [0.1, 1.0]$.

The categorical parameters include decay function $f \in \{\text{linear}, \text{exponential}, \text{concave}, \text{step}\}$, intensity encoding $e \in \{\text{binary}, \text{proportional}\}$, and color mapping $g \in \{\text{mono}, \text{twotone}, \text{gradient}, \text{intensity}\}$. Categorical choices are parameterized as logits over the option set and sampled via Gumbel-Softmax during training, enabling gradient flow through discrete decisions.

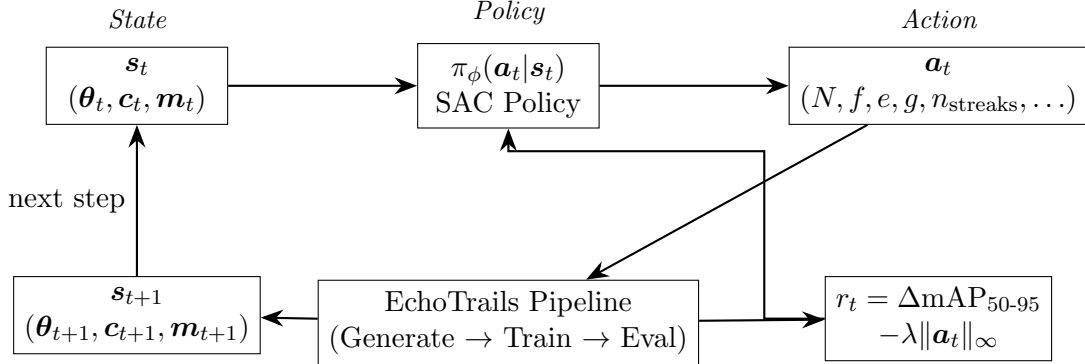


Figure 2: MDP formulation. The state $\mathbf{s}_t$ encodes the current parameter vector, scene context, and recent metrics. The SAC policy outputs a full generator configuration including trail length N , decay function f , intensity encoding e , color mapping g , and clutter parameters. The reward is the improvement in validation mAP_{50-95} with a regularization penalty discouraging extreme parameter settings.

2.3 Reward and Training

The reward r_t is the change in validation mAP_{50-95} between successive evaluation rounds, minus a regularization penalty $\lambda \|\mathbf{a}_t\|_\infty$ that discourages extreme parameter settings (e.g., trail lengths beyond $N = 25$ where the ablation study shows diminishing returns, or clutter intensities that saturate the image). Formally,

$$r_t = \text{mAP}_{50-95}^{(t+1)} - \text{mAP}_{50-95}^{(t)} - \lambda \|\mathbf{a}_t\|_\infty. \quad (1)$$

We adopt SAC [Haarnoja et al., 2018] for its sample efficiency with off-policy replay and its entropy-regularized exploration, which naturally encourages diverse parameter configurations during early training. The maximum entropy objective is well-suited to this problem because the generator parameter landscape is expected to contain multiple effective configurations, and premature convergence to a single setting would forfeit the diversity needed for robust detector training.

Each RL step requires executing the full EchoTrails pipeline (scene generation, polar-to-Cartesian conversion, echo trail rendering) followed by a 50-epoch YOLOv8n training run on 2,250 images and evaluation on 250 validation images. To amortize this cost, we use a replay buffer of 500 previous (state, action, reward, next-state) tuples and update the policy with 10 gradient steps per environment step.

2.4 Distinction from Prior Work

Our approach differs from prior work in three respects. First, unlike static domain randomization [Tobin et al., 2017, Tremblay et al., 2018], our policy adapts generator parameters *online* in response to downstream detection performance rather than sampling from a fixed distribution. Second, unlike Meta-Sim [Kar et al., 2019], which learns a single scene-generation distribution, our agent conditions on scene context (object density, clutter, speed tiers) to produce context-specific configurations. Third, unlike single-variable ablation, our RL formulation searches the *joint* parameter space, discovering interaction effects between trail rendering, clutter, and intensity parameters that isolated sweeps would miss.

3 Evaluation

We evaluate using the EchoTrails dataset infrastructure. The existing dataset comprises 10 scenes of 250 frames each (2,500 total), split 9:1 for training and validation. Each scene contains approximately 60 objects stratified across seven groups: stationary targets, slow/medium/fast movers originating from both edge and interior positions. All scenes use real Charleston harbor radar backgrounds (280 FURUNO-format frames) with a library of 5,106 filtered maritime target patches. Two conditions are tested: scenes with synthetic clutter (100 streaks per frame, intensity 200–255) and scenes without clutter.

3.1 Metrics

The primary metric is mAP_{50-95} (mean average precision averaged over IoU thresholds 0.50 to 0.95 in steps of 0.05), which is the standard COCO-style detection metric. We additionally report mAP_{50} , precision, recall, and F1 score. All metrics are computed separately for the two-class detection task (stationary vs. moving). We also report per-speed-tier detection rates (slow, medium, fast) to evaluate whether the RL-discovered parameters improve detection uniformly or favor particular motion regimes.

3.2 Baselines

We compare against four baselines.

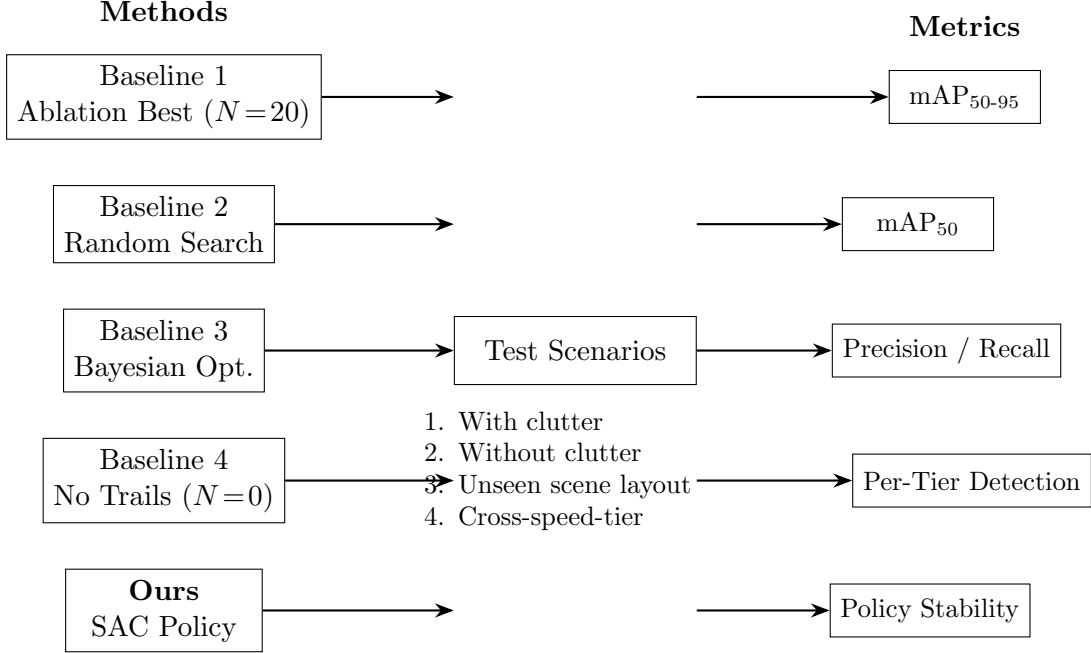


Figure 3: Evaluation pipeline. All methods are evaluated across four test scenarios spanning clutter conditions, unseen scene layouts, and cross-speed-tier generalization. Performance is measured by five complementary metrics.

The first baseline uses the best configuration found by the existing single-variable ablation study: $N = 20$ with linear decay, proportional intensity encoding, and twotone color mapping. This represents the current best-known parameter setting from manual experimentation. The second baseline replaces the RL policy with uniform random search over the full joint parameter space, sampling 100 configurations and reporting the best. The third baseline substitutes Bayesian optimization (Gaussian process with expected improvement acquisition) operating over the same parameter space, given the same computational budget as the RL agent. The fourth baseline uses no echo trails at all ($N = 0$, grayscale-to-RGB conversion only), establishing the lower bound of detection performance without temporal augmentation.

Evaluation follows four test scenarios of increasing difficulty. *With-clutter evaluation* tests under synthetic radar interference (100 streaks per frame). *Without-clutter evaluation* tests in clean conditions. *Unseen-scene-layout evaluation* generates new scene configurations with different object counts and placement seeds not seen during RL training. *Cross-speed-tier evaluation* tests whether parameters optimized for balanced speed distributions transfer to scenes dominated by a single speed tier (e.g., all fast movers at 12 px/frame). We additionally conduct ablation studies removing individual action dimensions (e.g., fixing N and optimizing only clutter and intensity parameters, or vice versa) to quantify the contribution of joint optimization.

4 Conclusion

We propose a reinforcement learning framework for jointly optimizing the parameters of the EchoTrails synthetic marine radar augmentation pipeline. By formulating the selection of trail length, decay function, intensity encoding, color mapping, clutter configuration, and per-object intensity and flicker settings as an MDP solved with SAC, our approach replaces expensive single-variable ablation with an adaptive closed-loop search over the full joint parameter space. The evaluation plan compares against the ablation-best configuration, random search, Bayesian optimization, and no-trail baselines across clutter conditions, unseen layouts, and cross-speed-tier scenarios. If successful, this work will demonstrate that RL-driven augmentation parameter optimization discovers configurations that outperform manual tuning, while revealing interaction effects in the generator parameter space that isolated ablation studies cannot capture.

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